RETAIL SALES PERFORMANCE ANALYSIS USING R

Nisha Patel (22207990)

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```r
# Load datasets
orders <- read_csv("D:/project_orders.csv")
products <- read_csv("D:/project_product_info.csv")

# Remove cancelled transactions
orders_clean <- orders %>%
  filter(!str_detect(InvoiceNo, "^C"))

# Join product descriptions
data <- orders_clean %>%
  left_join(products, by = "StockCode")

# Calculate revenue
data <- data %>%
  mutate(Revenue = Quantity * UnitPrice)

# Preview dataset
head(data)
```

```
## # A tibble: 6 x 9
##   InvoiceNo StockCode Quantity InvoiceDate    UnitPrice CustomerID Country
##   <chr>     <chr>        <dbl> <chr>              <dbl>      <dbl> <chr>
## 1 536365    85123A           6 1/12/2010 8:26      2.55      17850 United Kingd~
## 2 536365    71053            6 1/12/2010 8:26      3.39      17850 United Kingd~
## 3 536365    84406B           8 1/12/2010 8:26      2.75      17850 United Kingd~
## 4 536365    84029G           6 1/12/2010 8:26      3.39      17850 United Kingd~
## 5 536365    84029E           6 1/12/2010 8:26      3.39      17850 United Kingd~
## 6 536365    22752            2 1/12/2010 8:26      7.65      17850 United Kingd~
## # i 2 more variables: Description <chr>, Revenue <dbl>
```

Cleaning Code

```r
data <- orders %>%
  filter(!str_detect(InvoiceNo, "^C")) %>%   # remove cancelled invoices
  filter(Quantity > 0) %>%                   # remove returns
  filter(UnitPrice > 0) %>%                  # remove free items
  left_join(products, by = "StockCode") %>%
  mutate(Revenue = Quantity * UnitPrice)
```

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#template to answer the questions

Q1: Product Revenue Analysis

Explanation

This is an analysis of product performance, which involves the calculation of the total revenue per product. The formula of the determination of revenue is Quantity x UnitPrice. The rank of products is used to determine the top ten and the bottom ten performers. Also, the contribution that the three top-selling products make to the revenue is calculated to determine the patterns of revenue concentration. ### Code

```r
# Revenue per product
product_revenue <- data %>%
  group_by(Description) %>%
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE)) %>%
  arrange(desc(TotalRevenue))

# Top 10 products
top10_products <- product_revenue %>%
  slice_max(TotalRevenue, n = 10)

# Bottom 10 products
bottom10_products <- product_revenue %>%
  slice_min(TotalRevenue, n = 10)

# Top 3 revenue contribution
top3 <- product_revenue %>%
  slice_max(TotalRevenue, n = 3)

top3_percentage <- sum(top3$TotalRevenue) /
  sum(product_revenue$TotalRevenue) * 100
```

```r
# Display tables
kable(top10_products, caption = "Top 10 Products by Revenue")
```

Table 1: Top 10 Products by Revenue

Description	TotalRevenue
DOTCOM POSTAGE	206248.77

Description	TotalRevenue
REGENCY CAKESTAND 3 TIER	174484.74
PAPER CRAFT , LITTLE BIRDIE	168469.60
WHITE HANGING HEART T-LIGHT HOLDER	106471.28
mailout	99568.04
PARTY BUNTING	99504.33
JUMBO BAG RED RETROSPOT	94340.05
MEDIUM CERAMIC TOP STORAGE JAR	81700.92
Manual	78112.82
POSTAGE	78101.88

Interpretation

The top 10 products that have contributed to the highest total revenue are shown in this figure. The products with the most revenue of about 206,248.77 are DOTCOM POSTAGE, then comes REGENCY CAKESTAND 3 TIER with 174,484.74 and PAPER CRAFT, LITTLE BIRDIE with 168,469.60. These products play a major role in the total sales performance in the dataset under analysis.

```
kable(bottom10_products, caption = "Bottom 10 Products by Revenue")
```

Table 2: Bottom 10 Products by Revenue

Description	TotalRevenue
PADS TO MATCH ALL CUSHIONS	0.003
VINTAGE BLUE TINSEL REEL	0.840
PINK CRYSTAL GUITAR PHONE CHARM	0.850
CAT WITH SUNGLASSES BLANK CARD	0.950
HAPPY BIRTHDAY CARD TEDDY/CAKE	0.950
BLACKCHRISTMAS TREE 30CM	1.250
DUSTY PINK CHRISTMAS TREE 30CM	1.250
ETCHED GLASS STAR TREE DECORATION	1.250
ORANGE FELT VASE + FLOWERS	1.250
PACK 4 FLOWER/BUTTERFLY PATCHES	1.250

Interpretation

This figure shows the top ten products that make the least revenue in the dataset (Wang and Gu, 2022). PADS TO MATCH ALL CUSHIONS has the lowest revenue of about 0.003, whereas items like VINTAGE BLUE TINSEL REEL and PINK CRYSTAL GUITAR PHONE CHARM have less than 1.00 revenue, meaning that there are a very low number of transactions where they are in demand.

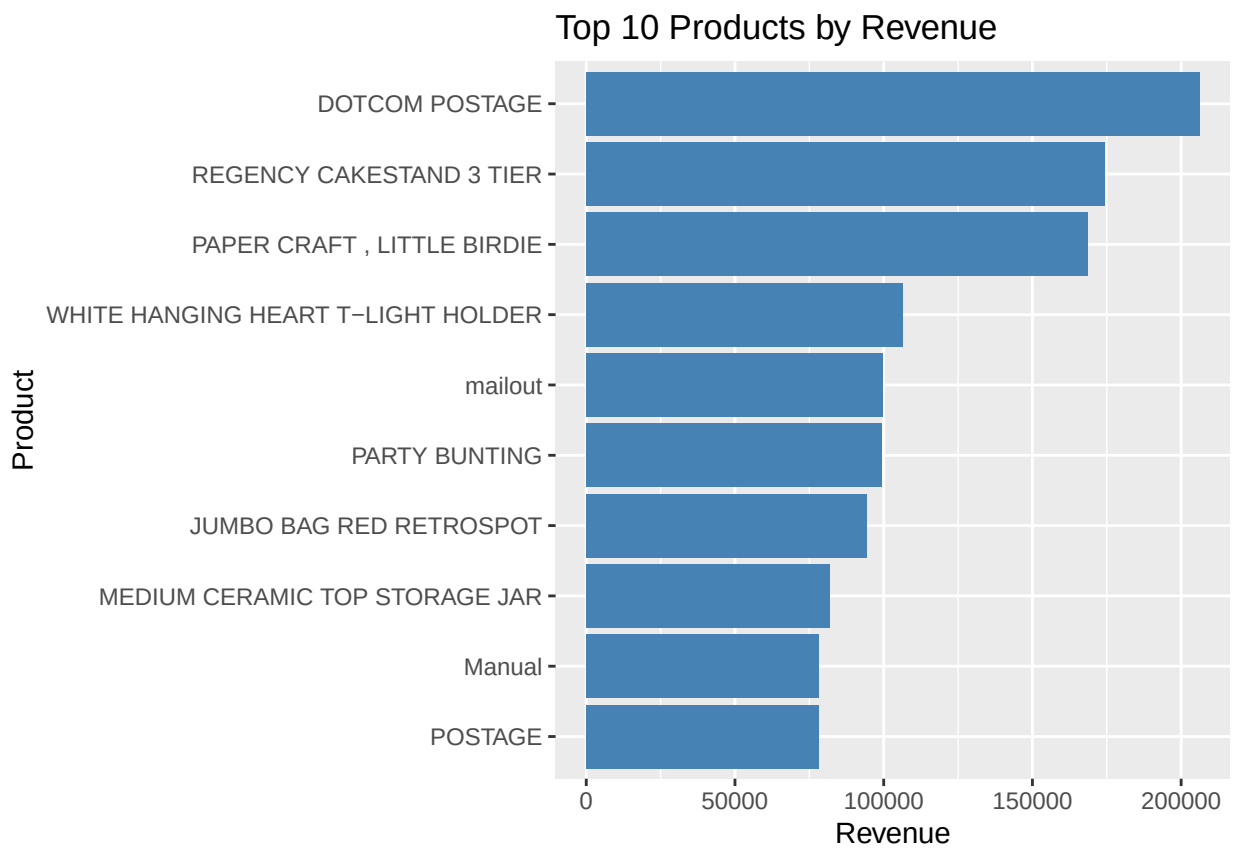
```
# Display percentage
top3_percentage
```

```
## [1] 5.148771
```

Interpretation

This figure shows the revenue contribution of the 3 highest performing products in the dataset. The contribution of DOTCOM POSTage, Regency cakestand 3 tier and Paper craft is around 5.15% of the total revenue. It means that a few products are selling a significant share of the total sales revenue. ### Visualisation

```r
ggplot(top10_products, aes(x=reorder(Description,TotalRevenue),
                           y=TotalRevenue)) +
  geom_bar(stat="identity", fill="steelblue") +
  coord_flip() +
  labs(title="Top 10 Products by Revenue",
       x="Product",
       y="Revenue")
```



Interpretation

This figure represents the ten leading products in terms of revenue in a horizontal bar chart. About 206,248 comes first with DOTCOM POSTAGE, and then comes REGENCY CAKESTAND 3 TIER, which comes in at number 174,484. Other products, such as WHITEN Hanging heart T-light holder and Party Bunting, make over 90,000 in revenues, which means that they are in high demand in various transactions (Sajawal et al. 2022).

Q2: High Value Customer Analysis

This analysis determines customers who produce the best revenue due to their purchases. The cumulative expenditures on the individual customers are computed as a sum of transaction revenues. The top five highest spending customers are identified and their buying behaviour analysed. The amount of products bought by these customers is analysed in order to know their preferences and customer value.

```r
# Total spending per customer
customer_spending <- data %>%
  filter(!is.na(CustomerID)) %>%
  group_by(CustomerID) %>%
  summarise(TotalSpending = sum(Revenue, na.rm = TRUE)) %>%
  arrange(desc(TotalSpending))

# Top 5 customers
top5_customers <- customer_spending %>%
  slice_max(TotalSpending, n = 5)

# Display table
knitr::kable(top5_customers, caption = "Top 5 Customers by Total Spending")
```

Table 3: Top 5 Customers by Total Spending

CustomerID	TotalSpending
14646	280206.0
18102	259657.3
17450	194550.8
16446	168472.5
14911	143825.1

Interpretation

The top five customers in terms of contribution to revenue are observed in this figure. Customer number 14646 has the largest expenditure of about 280,206, and customer number 18102 comes second with 259,657. Other customers, such as 17450, 16446 and 14911, provide a high purchasing value, and it means that high-value customers are very engaged (Falatouri et al. 2022).

Customer Product Preference

```r
top_customer_products <- data %>%
  filter(CustomerID %in% top5_customers$CustomerID) %>%
  group_by(CustomerID, Description) %>%
  summarise(TotalQuantity = sum(Quantity)) %>%
  arrange(desc(TotalQuantity))

head(top_customer_products)
```

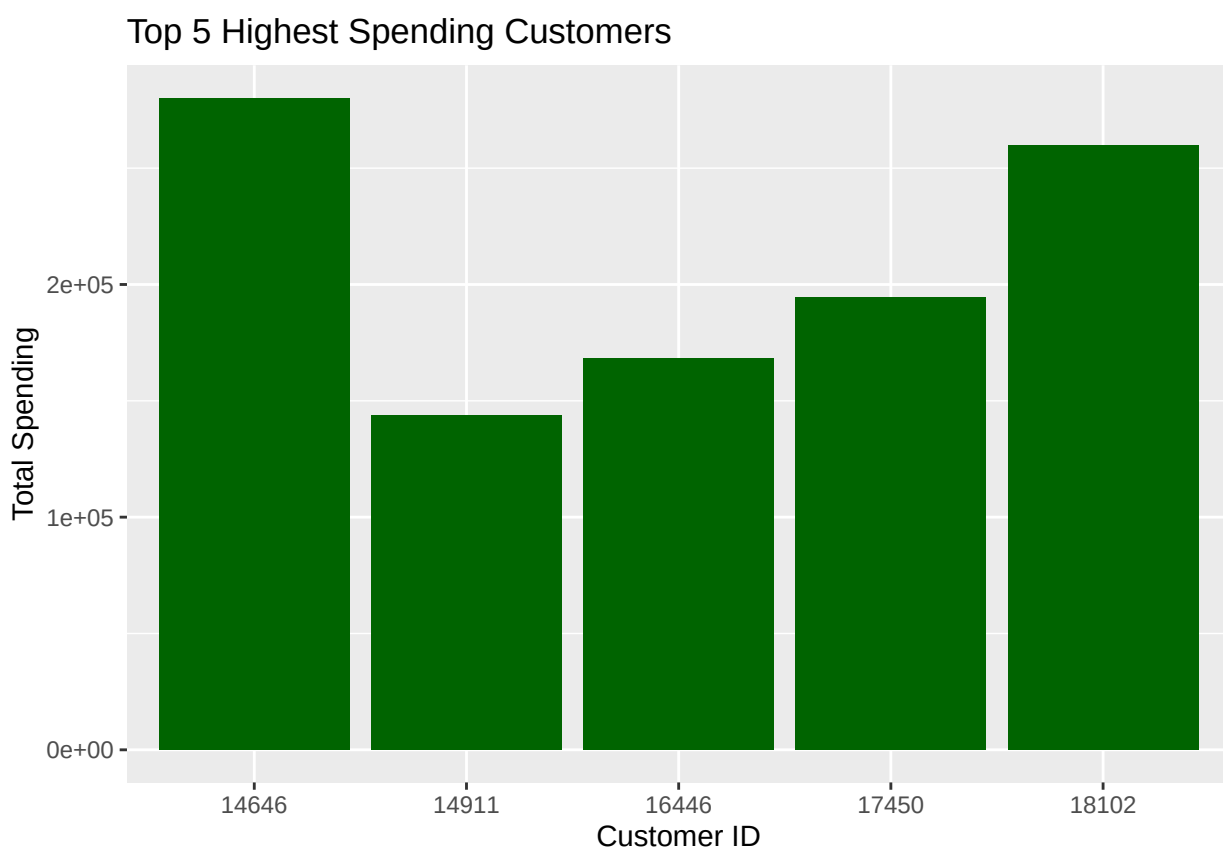
```
## # A tibble: 6 x 3
## # Groups:   CustomerID [4]
##   CustomerID Description                 TotalQuantity
##        <dbl> <chr>                               <dbl>
## 1      16446 PAPER CRAFT , LITTLE BIRDIE         80995
## 2      18102 CREAM HEART CARD HOLDER              5946
## 3      17450 HEART OF WICKER SMALL                5286
## 4      14646 RABBIT NIGHT LIGHT                   4801
## 5      14646 SPACEBOY LUNCH BOX                   4492
## 6      17450 HEART OF WICKER LARGE                4282
```

Interpretation

This figure highlights the most bought products by the best customers. The highest level of product quantity that is purchased is Customer 16446 buying PAPER CRAFT, LITTLE BIRDIE in a quantity of 80,995. There are also other products like CREAM HEART CARD HOLDER and HEART of Wicker small that record quantities of more than 5,000.

Visualisation

```
ggplot(top5_customers,
       aes(x=factor(CustomerID), y=TotalSpending)) +
  geom_bar(stat="identity", fill="darkgreen") +
  labs(title="Top 5 Highest Spending Customers",
       x="Customer ID",
       y="Total Spending")
```



Interpretation

The comparisons of the five top customers in terms of value are presented as a bar chart. The biggest customer is 14646, with an expenditure of about 280,206, preceded by 18102 with a 259,657 expenditure. The rest of the customers portray a range of 143,825 to 194,550 amounts spent by the customers, which attests to the high contribution being made by the loyal customers.

Q3: Global Market Analysis

Explanation

This comparison analyses the demand for its product in the various countries by summing all the revenue per country. The transactions are classified by country to establish the total contribution of the market. Also, the highest-selling product in each country is determined in terms of total quantity sold, which serves to indicate the buying behaviour in the region and the variation between markets.

```r
# Revenue by country
country_revenue <- data %>%
  group_by(Country) %>%
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE)) %>%
  arrange(desc(TotalRevenue))

# Display table
knitr::kable(country_revenue, caption = "Total Revenue by Country")
```

Table 4: Total Revenue by Country

Country	TotalRevenue
United Kingdom	9025222.08
Netherlands	285446.34
EIRE	283453.96
Germany	228867.14
France	209715.11
Australia	138521.31
Spain	61577.11
Switzerland	57089.90
Belgium	41196.34
Sweden	38378.33
Japan	37416.37
Norway	36165.44
Portugal	33747.10
Finland	22546.08
Singapore	21279.29
Channel Islands	20450.44
Denmark	18955.34
Italy	17483.24
Hong Kong	15691.80
Cyprus	13590.38
Austria	10198.68
Israel	8135.26
Poland	7334.65
Greece	4760.52
Unspecified	4749.79
Iceland	4310.00
Canada	3666.38
USA	3580.39
Malta	2725.59
United Arab Emirates	1902.28
Lebanon	1693.88

Country	TotalRevenue
Lithuania	1661.06
European Community	1300.25
Brazil	1143.60
RSA	1002.31
Czech Republic	826.74
Bahrain	754.14
Saudi Arabia	145.92

Interpretation

The overall revenue by each country is summed up in the dataset. Poland has revenue of some 7,334.65, and Greece and Unspecified have revenues of 4,760.52 and 4,749.79, respectively. The table points out disparities in the market performance of the regions available in various global sites.

Most popular product per country

```
country_products <- data %>%
  group_by(Country, Description) %>%
  summarise(TotalQuantity = sum(Quantity)) %>%
  slice_max(TotalQuantity, n=1)

head(country_products)
```

```
## # A tibble: 6 x 3
## # Groups:   Country [5]
##   Country   Description                       TotalQuantity
##   <chr>     <chr>                                     <dbl>
## 1 Australia MINI PAINT SET VINTAGE                     2952
## 2 Austria   SET 12 KIDS COLOUR  CHALK STICKS           288
## 3 Bahrain   ICE CREAM SUNDAE LIP GLOSS                  96
## 4 Belgium   PACK OF 72 RETROSPOT CAKE CASES            480
## 5 Brazil    DOLLY GIRL LUNCH BOX                        24
## 6 Brazil    GREEN REGENCY TEACUP AND SAUCER             24
```

Interpretation

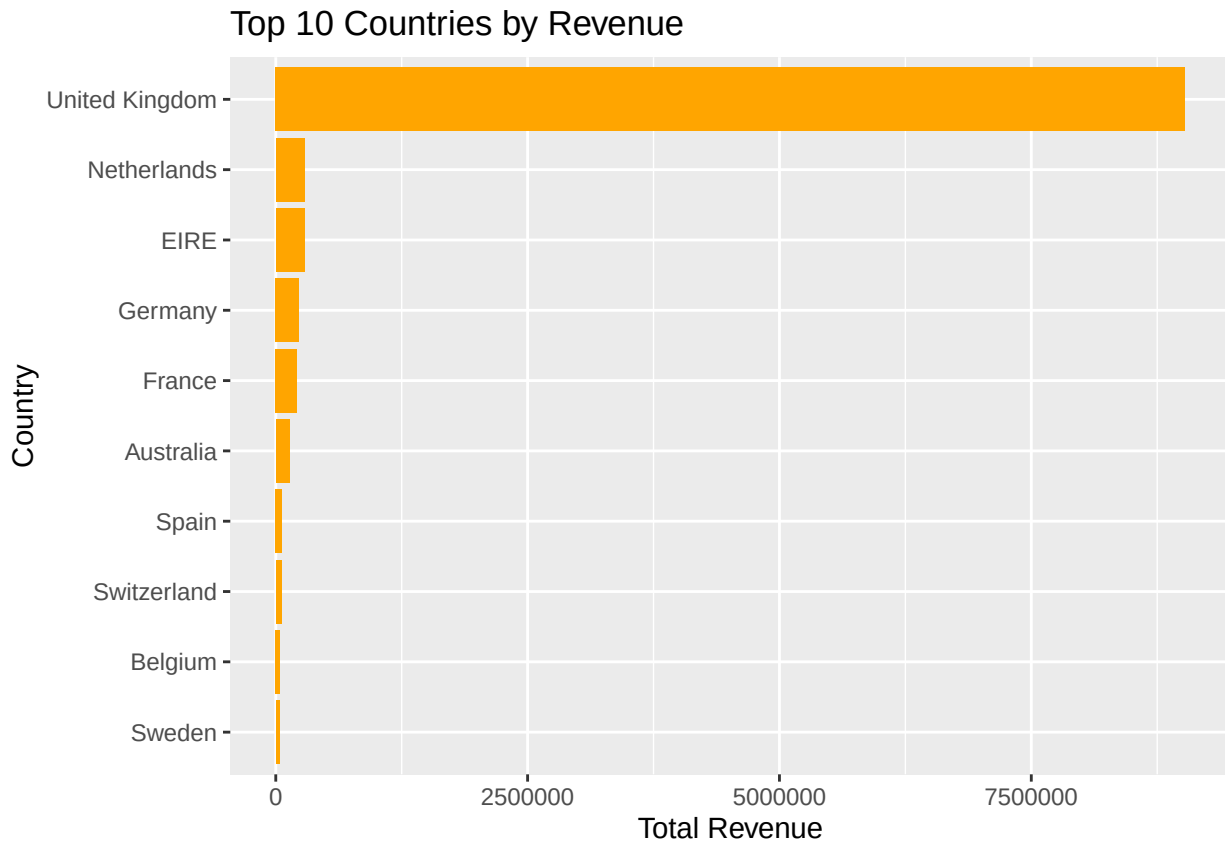
The most sold product in each country in terms of total quantity sold. Australia has MINI PAINT SET VINTAGE and 2,952 units sold, whereas Austria has SET 12 KIDS COLOUR CHALK STICKS with 288 units, which reflects regional product preferences in the international markets.

Visualisation

```
top_countries <- country_revenue %>%
  slice_max(TotalRevenue, n=10)

ggplot(top_countries,
```

```
     aes(x=reorder(Country,TotalRevenue),
         y=TotalRevenue)) +
geom_bar(stat="identity", fill="orange") +
coord_flip() +
labs(title="Top 10 Countries by Revenue",
     x="Country",
     y="Total Revenue")
```



Interpretation

The visualisation of the top ten countries producing the highest total revenue. The United Kingdom leads with a revenue of more than 9,000,000,000, which is way higher than other nations. The revenues of the Netherlands, EIRE, and Germany are relatively small, which proves the high level of concentration of sales on the British market.

Q4: Temporal Analysis and Sales Trends

Explanation

This is an analysis of the change in revenue with time using transaction dates. The InvoiceDate variable is used to obtain the month and year, which are used to compute monthly revenue amounts. Time-series

trend is analysed to determine the high sales periods. Products that keep recurring in the lowest monthly revenues are also listed to be evaluated on their performance.

```
# Convert date
data <- data %>%
  mutate(
    InvoiceDate = dmy_hms(InvoiceDate),
    Year = year(InvoiceDate),
    Month = month(InvoiceDate, label = TRUE)
  )

# Monthly revenue
monthly_sales <- data %>%
  group_by(Year, Month) %>%
  summarise(MonthlyRevenue = sum(Revenue, na.rm = TRUE)) %>%
  arrange(Year, Month)

# Display table
knitr::kable(monthly_sales, caption = "Monthly Revenue Trend")
```

Table 5: Monthly Revenue Trend

Year	Month	MonthlyRevenue
2020	Jan	691364.6
2020	Feb	523631.9
2020	Mar	717639.4
2020	Apr	537808.6
2020	May	770536.0
2020	Jun	761739.9
2020	Jul	719221.2
2020	Aug	759138.4
2020	Sep	1058590.2
2020	Oct	1154979.3
2020	Nov	1509496.3
2020	Dec	1462538.8

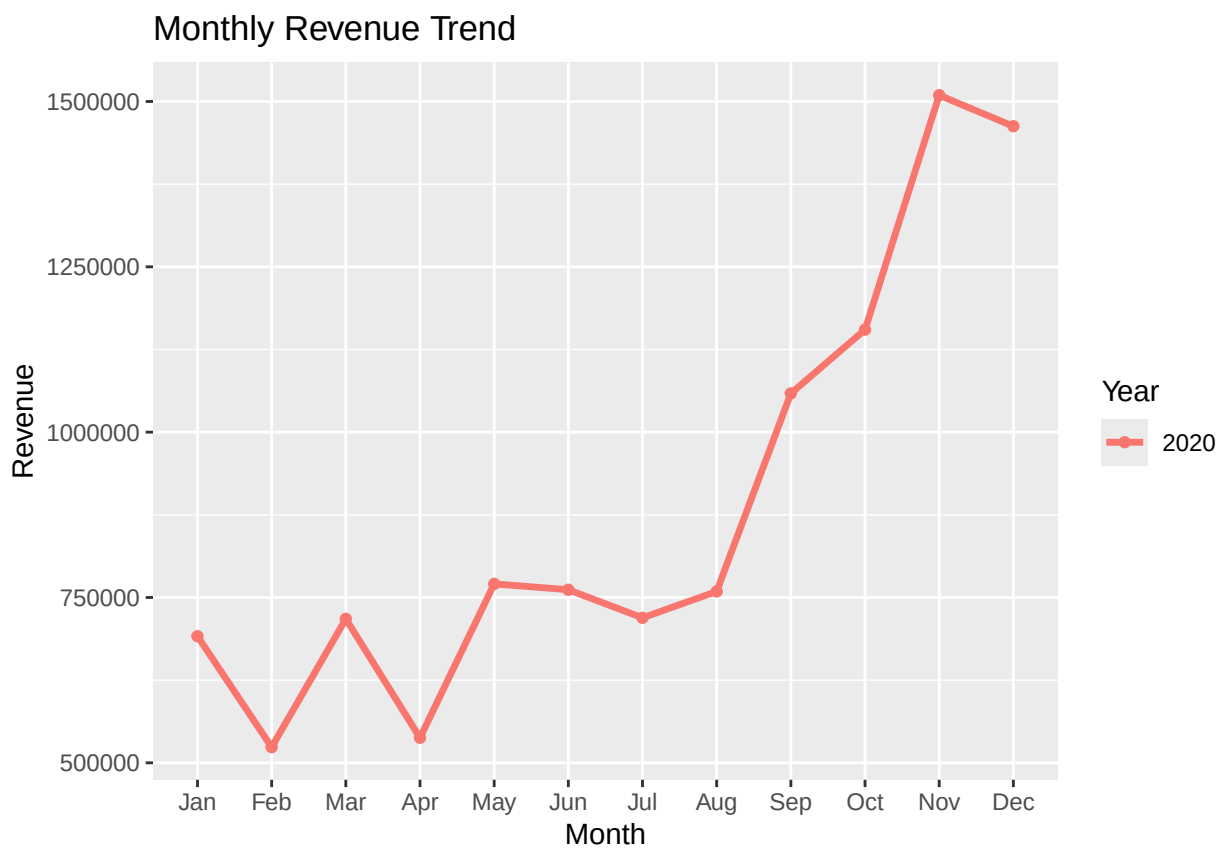
Interpretation

The values of monthly revenues in 2020 are given in this figure. The starting point of the revenue is around 691,364.6 in January and varies during the year. The highest sales are recorded in November at about 1,509,496.3 and in December, at 1,462,538.8, which is more purchasing activity towards the end of the year (Suszynski et al. 2022).

Visualisation

```
ggplot(monthly_sales,
       aes(x=Month, y=MonthlyRevenue, group=Year, colour=factor(Year))) +
  geom_line(size=1.2) +
  geom_point() +
  labs(title="Monthly Revenue Trend",
```

```
       x="Month",
       y="Revenue",
       colour="Year")
```



Interpretation

The year 2020 shows the monthly revenue trends on a line chart, as shown in this figure. The revenue is slow to grow, with approximately 523,631 in February and more than 1,150,000 in October. The peak is recorded in the month of November and at about 1,509,496, which means high season sales growth in the last months.

Lowest selling products across months

```
lowest_products <- data %>%
  group_by(Month, Description) %>%
  summarise(MonthRevenue = sum(Revenue), .groups="drop") %>%
  group_by(Month) %>%
  slice_min(MonthRevenue, n=5) %>%      # lowest 5 products per month
  ungroup()

# Count how often products appear as lowest sellers
consistent_low <- lowest_products %>%
  count(Description, sort=TRUE)
```

```r
head(consistent_low,10)
```

```
## # A tibble: 10 x 2
##    Description                          n
##    <chr>                            <int>
##  1 "LETTER \"F\" BLING KEY RING"        3
##  2 "PADS TO MATCH ALL CUSHIONS"         3
##  3 "PET MUG, GOLDFISH"                  3
##  4 "ASSORTED COLOUR SUCTION CUP HOOK"   2
##  5 "BLUE FLOCK GLASS CANDLEHOLDER"      2
##  6 "CD WALL TIDY BLUE OFFICE"           2
##  7 "FAN BLACK FRAME"                    2
##  8 "GLOW IN DARK DOLPHINS"              2
##  9 "HARDMAN MUG 3 ASSORTED"             2
## 10 "LETTER \"C\" BLING KEY RING"        2
```

Interpretation

This figure marks the products that are constantly listed as the lowest-selling products in several months. Among the lowest performers are such products as LETTER "F" BLING KEY RING, PADS TO MATCH ALL CUSHIONS, PET MUG, three times, which means that the demand is low as well as sales contribution.

References

Wang, C.H. and Gu, Y.W., 2022. Sales forecasting, market analysis, and performance assessment for US retail firms: a business analytics perspective. Applied sciences, 12(17), p.8480.

Sajawal, M., Usman, S., Alshaikh, H.S., Hayat, A. and Ashraf, M.U., 2022. A predictive analysis of retail sales forecasting using machine learning techniques. Lahore Garrison University Research Journal of Computer Science and Information Technology, 6(04), pp.33-45.

Falatouri, T., Darbanian, F., Brandtner, P. and Udokwu, C., 2022. Predictive analytics for demand forecasting–a comparison of SARIMA and LSTM in retail SCM. Procedia Computer Science, 200, pp.993-1003.

Suszynski, C., Dao, M.T.H. and Minh, D.T., 2022. Cost optimization for R-logistics operations at foreign supermarkets in Vietnam. Case of AEON and Lotte. International Journal of Multidisciplinary Research and Growth Evaluation, 3(1), pp.210-216.

I confirm that: - I understand all code submitted. - I can explain my solution if required.